

Experimental Report on Linear Regression using Gradient Descent

1. Dataset Selection and Processing

The dataset used for this experiment is 'data.csv', which contains two columns: 'X' and 'Y'. The data was preprocessed by filling missing values with the mean, removing duplicates, and filtering outliers using the Interquartile Range (IQR) method.

2. Implementation of Gradient Descent

The gradient descent method was implemented to perform simple linear regression. The initial weights (w , b) were set to 0, and the learning rate was set to 0.01. The algorithm iteratively updated the weights over 1000 iterations to minimize the cost function.

3. Analysis and Discussion of Regression Results

The regression line was plotted against the training data, showing a good fit. The final parameters obtained were $w = 2.0099$ and $b = 2.7066$. The cost function decreased steadily over the iterations, indicating convergence.

4. Interpretation of Visualization Charts

The 'training_results.png' shows the training data and the regression line. The 'cost_function_plot.png' illustrates the decrease in the cost function over iterations, demonstrating the effectiveness of the gradient descent. The '3D_contour_plot.png' and '2D_contour_plot.png' visualize the cost function's landscape, providing insights into the parameter space and the optimization process.